

Interacting Multipoles of Rank 1 and 4:

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_{\alpha}\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{105} \cdot \left[\begin{array}{l} -945 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot \delta(\delta, \epsilon) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\delta} \cdot \delta(\gamma, \epsilon) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\epsilon} \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta(\beta, \epsilon) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\epsilon} \cdot \delta(\beta, \delta) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta(\beta, \gamma) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta(\alpha, \epsilon) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\epsilon} \cdot \delta(\alpha, \delta) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta(\alpha, \gamma) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta(\alpha, \beta) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_{\alpha} \cdot \delta(\beta, \gamma) \cdot \delta(\delta, \epsilon) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_{\alpha} \cdot \delta(\beta, \delta) \cdot \delta(\gamma, \epsilon) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_{\alpha} \cdot \delta(\beta, \epsilon) \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_{\beta} \cdot \delta(\alpha, \gamma) \cdot \delta(\delta, \epsilon) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_{\beta} \cdot \delta(\alpha, \delta) \cdot \delta(\gamma, \epsilon) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_{\beta} \cdot \delta(\alpha, \epsilon) \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_{\gamma} \cdot \delta(\alpha, \beta) \cdot \delta(\delta, \epsilon) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_{\gamma} \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \epsilon) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_{\gamma} \cdot \delta(\alpha, \epsilon) \cdot \delta(\beta, \delta) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_{\delta} \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \epsilon) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_{\delta} \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \epsilon) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_{\delta} \cdot \delta(\alpha, \epsilon) \cdot \delta(\beta, \gamma) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_{\epsilon} \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_{\epsilon} \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \delta) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_{\epsilon} \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \gamma) \cdot \mu_{\alpha} \cdot \Phi_{\beta\gamma\delta\epsilon} \end{array} \right] \quad (1)$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_\alpha\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{105} \cdot$$

$(2b)$

Simplify R:

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_{\alpha}\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{105} \cdot \left[\begin{array}{l} -945 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Phi_{zzzzz} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Phi_{zz\delta\delta} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Phi_{zz\gamma\gamma} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Phi_{zz\gamma\gamma} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Phi_{zz\beta\beta} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Phi_{zz\beta\beta} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Phi_{zz\beta\beta} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_{\alpha} \cdot \Phi_{zzz\alpha} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_{\alpha} \cdot \Phi_{zzz\alpha} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_{\alpha} \cdot \Phi_{zzz\alpha} \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_{\alpha} \cdot \Phi_{zzz\alpha} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_z \cdot \Phi_{\beta\beta\delta\delta} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_z \cdot \Phi_{\beta\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_z \cdot \Phi_{\beta\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\delta\delta} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\delta\delta} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\beta\beta} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\beta\beta} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\beta\beta} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\beta\beta} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_z \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\beta\beta} \end{array} \right] \quad (3a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-11} \cdot -\frac{1}{105} \cdot \left[\begin{array}{l} +105 \cdot R_z^5 \cdot \mu_{\alpha} \cdot \Phi_{zzz\alpha} \\ +105 \cdot R_z^5 \cdot \mu_{\alpha} \cdot \Phi_{zzz\alpha} \\ +105 \cdot R_z^5 \cdot \mu_{\alpha} \cdot \Phi_{zzz\alpha} \\ +105 \cdot R_z^5 \cdot \mu_{\alpha} \cdot \Phi_{zzz\alpha} \\ -15 \cdot R_z^5 \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_{\alpha} \cdot \Phi_{z\alpha\beta\beta} \end{array} \right] \quad (3b)$$

Simplify Multipoles:

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_\alpha\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{105} \cdot \left[\begin{array}{l} +105 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yzzzz} \\ +105 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yzzzz} \\ +105 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yzzzz} \\ +105 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yzzzz} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \end{array} \right] \quad (4a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-11} \cdot -\frac{1}{105} \cdot \left[\begin{array}{l} -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \end{array} \right] \quad (4b)$$

Exploit Traceless Conditions:

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_\alpha\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{105} \cdot \begin{bmatrix} -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \end{bmatrix} \quad (5a)$$

$$\xRightarrow{\text{cleaned up}} R_z^{-11} \cdot -\frac{1}{105} \cdot \begin{bmatrix} -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \end{bmatrix} \quad (5b)$$

Reduce Indices:

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_{\alpha}\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{105} \cdot \begin{bmatrix} -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \end{bmatrix} \quad (6a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-11} \cdot -\frac{1}{105} \cdot \begin{bmatrix} -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \\ -15 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \end{bmatrix} \quad (6b)$$

Add up:

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_{\alpha}\Phi_{\beta\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{105} \cdot \begin{bmatrix} -30 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ -60 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ -90 \cdot R_z^5 \cdot \mu_y \cdot \Phi_{yz\beta\beta} \end{bmatrix} \quad (7)$$

Combining everything:

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_{\alpha}\Phi_{\beta\gamma\delta\epsilon} = \begin{bmatrix} +\frac{2}{7} \cdot R_z^{-6} \cdot \mu_y \cdot \Phi_{yz\delta\delta} \\ +\frac{4}{7} \cdot R_z^{-6} \cdot \mu_y \cdot \Phi_{yz\gamma\gamma} \\ +\frac{6}{7} \cdot R_z^{-6} \cdot \mu_y \cdot \Phi_{yz\beta\beta} \end{bmatrix} \quad (8)$$